Unnumbered continuation: compactly supported corrections to a common fixed region. Retain the actual data and the exact maps of C.81, C.92 and (CFT.1): the original closed disk $\overline{D}$, its interior O, the compatible metric d, the free proper closed discrete operator group Q, the boundary-fixed centralizer $\mathscr{C}_Q$ with a transitive actual subgroup, the covering $\pi : O \to N = O/Q$, and the specified points $x \in O$, $p = \pi x$. Write $\mathscr{F} = \mathscr{C}_Q(x)$. The quotient N is noncompact in this note. All compositions below act first by the rightmost factor.

Let (P, u_0) be a compact metric based space and let $u \mapsto h_u$ be a continuous map to $\mathscr{F}$ in the original uniform map-and-inverse topology, with $h_{u_0} = 1$. For every compact subpolyhedron $X \subset N$ containing p, the construction below gives a based deformation in the original group,

$$\begin{aligned} H : P \times [0,1] \longrightarrow \mathscr{F}, \qquad H_{u,0} = h_u, \qquad H_{u_0,t} = 1, \\ H_{u,1}|_{\pi^{-1}X} = \mathrm{id}, \qquad H_{u,t} = G_{u,t}^{-1} h_u, \qquad H_{u,t}^{-1} = h_u^{-1} G_{u,t}, \end{aligned} \tag{IFR.1}$$

where $G_{u,0} = G_{u_0,t} = 1$ and every $G_{u,t}^{\pm 1}$ is the identity off the inverse image of one fixed compact subset of N. Every original deck label is retained. The terminal descents also come with an explicit ordinary isotopy relative to X from the identity. This last property permits compatible successive stages for any finite nested sequence of compact polyhedra beginning with a closed disk neighborhood of p.

The exact homeomorphism-group topology downstairs. For a surface M without boundary, the compact-open topology of homeomorphisms equals the compact-open topology of maps and inverses. Here is the comparison needed in this construction. Let $a \mapsto f_a$ be any family continuous for compact-open convergence of maps. Evaluation is jointly continuous, since M is locally compact. Fix (a_*, y_*) and put $z_* = f_{a_*}^{-1}(y_*)$. Inside any prescribed neighborhood of z_* choose a closed coordinate disk D_0 with $z_* \in \operatorname{int} D_0$. Choose a connected open coordinate disk V about y_* with compact closure in $f_{a_*}(\operatorname{int} D_0)$. For a in a neighborhood of a_*, the compact-open conditions give

$$f_a(\partial D_0) \cap \overline{V} = \varnothing, \qquad f_a(z_*) \in V. \tag{IFR.2}$$

The two relatively open sets $V \cap f_a(\operatorname{int} D_0)$ and $V \setminus f_a(D_0)$ partition V, because its points avoid $f_a(\partial D_0)$. The first is nonempty by (IFR.2). Connectedness therefore gives $V \subset f_a(\operatorname{int} D_0)$. Thus $f_a^{-1}(V) \subset D_0$, proving joint inverse continuity. The compact-open exponential law now proves continuity of inversion. The two identity maps between the two stated topologies are continuous: one forgets inverse coordinates, and the other uses this inversion map. Composition is continuous by joint evaluation and the same exponential law. These are the group topology and exact comparison used below.

Use the quotient triangulation from C.82, subdivided to make p a vertex. The surface is separable: the open covering map π sends a countable basis of O to a countable basis of N. Consequently the cited PL-surface results apply to this actual surface, with its compact subpolyhedra. Its boundary is empty, so the "proper embedding" condition in the source theorem imposes no additional condition on the embeddings considered here.

The supported local extension theorem in its exact form. For compact subpolyhedra $K \subset X \subset M$, write $\mathcal{E}_K(X, M)$ for embeddings restricting to the identity on K. The primary statement [1, Theorem 2.1, p. 2] says that for each such embedding f and every neighborhood U of $f(X)$ there are a neighborhood $\mathcal{V}_f$ of f and a continuous map

$$\phi_f : \mathcal{V}_f \longrightarrow \mathrm{Homeo}_{K \cup (M \setminus U)}(M)_0, \qquad \phi_f(g) f = g, \qquad \phi_f(f) = \mathrm{id}_M . \tag{IFR.3}$$

The subscript 0 denotes the identity component. The identities in (IFR.3) specify the actual extension and its support. The topology comparison above supplies continuous inverses as well.

We give the finite parametric extension construction explicitly. Let $e : P \times [0,1] \to \mathcal{E}_K(X, M)$ be a continuous family with $e_{u,0} = i_X$ and $e_{u_0,t} = i_X$. This notation will be applied below to the restrictions of a specified family of actual homeomorphisms. The trace $\{e_{u,t}(z) : u \in P, t \in$

$[0,1], z \in X\}$ is compact. Choose one relatively compact open $U \subset M$ containing this trace. For every reference embedding in the compact parameter image, apply (IFR.3) with this same U. The inverse images of finitely many resulting neighborhoods cover $P \times [0,1]$.

Give this compact parameter space the maximum product metric, and choose a Lebesgue number $\delta > 0$ for that finite open cover. Choose a time subdivision $0 = a_0 < \cdots < a_n = 1$ with successive lengths less than $\delta/3$. Choose finitely many points $v_i \in P$ whose open $\delta/8$ balls cover P, and put

$$\begin{aligned} U_i &= B_P(v_i, \delta/3), \qquad \beta_i(u) = \max\{0, \delta/4 - d_P(u, v_i)\}, \\ \alpha_i(u) &= \frac{\beta_i(u)}{\sum_j \beta_j(u)}, \qquad \sum_i \alpha_i = 1, \qquad \operatorname{supp} \alpha_i \subset U_i. \end{aligned} \tag{IFR.4}$$

The denominator is positive by the chosen cover. Each support is contained in the closed $\delta/4$ ball, which lies inside U_i. For each time strip $[a,b] = [a_{k-1}, a_k]$, the set $\overline{U}_i \times [a,b]$ has diameter less than δ. It is therefore contained in the inverse image of one extension chart. Denote that chart's map by $\phi_{i,k}$.

For $t \in [a,b]$ set

$$\begin{gathered} t_0(u,t) = a, \qquad t_i(u,t) = a + (t-a) \sum_{j \le i} \alpha_j(u), \qquad t_m(u,t) = t, \\ \Gamma_{i,k}(u,t) = \phi_{i,k}(e_{u,t_i})\, \phi_{i,k}(e_{u,t_{i-1}})^{-1} \quad (u \in U_i), \\ \Gamma_{i,k}(u,t) = \mathrm{id}_M \quad (u \notin \operatorname{supp} \alpha_i), \\ \Gamma_{i,k}(u,t) e_{u,t_{i-1}} = e_{u,t_i}. \end{gathered} \tag{IFR.5}$$

Here m is the number of the sets U_i. Both arguments in the chart formula lie in its domain. Where $\alpha_i = 0$ their time arguments coincide, making that formula the identity. The two definitions therefore agree on the overlap of the open sets U_i and $P \setminus \operatorname{supp} \alpha_i$, which cover P. This proves continuity of the pasted maps and of their explicit inverses $\phi_{i,k}(e_{u,t_{i-1}})\phi_{i,k}(e_{u,t_i})^{-1}$. The last identity in (IFR.5) follows by first applying the inverse extension to $e_{u,t_{i-1}}$ to recover its reference embedding, then applying the other extension.

Starting with $g_{u,0} = 1$, define successively on the time strips

$$\begin{aligned} g_{u,t} &= \Gamma_{m,k}(u,t) \cdots \Gamma_{1,k}(u,t) g_{u,a}, \\ g_{u,t}^{-1} &= g_{u,a}^{-1} \Gamma_{1,k}(u,t)^{-1} \cdots \Gamma_{m,k}(u,t)^{-1}, \\ g_{u,t}|_X = e_{u,t}, \qquad & g_{u_0,t} = 1, \qquad g_{u,t}|_{K \cup (M \setminus U)} = \mathrm{id}\,. \end{aligned} \tag{IFR.6}$$

At $t = a$ all factors $\Gamma_{i,k}$ are the identity, so consecutive strip definitions agree exactly. Finite pasting proves joint continuity. Induction using the last equation of (IFR.5) proves the restriction identity. At $u = u_0$ both embedding arguments are the same inclusion, so every factor is the identity, even when the chart's reference embedding is different. The support and fixed-set assertions follow from (IFR.3) factor by factor. Thus (IFR.6) gives the supported extension with its inverse and based parameter behavior, without an unwritten choice of a section of a parameter bundle.

The initial ordinary isotopy and the pointed-plane exception. Let $b_u = D(h_u)$, using the exact descent (CFT.1). These maps and their inverses depend continuously on u, fix p, and belong to the pointed identity path component by C.92.

For nontrivial Q, the quotient N cannot be the plane. Indeed a plane coordinate gives paths from p to every point by straight segments. Lifting those paths from x gives a continuous section $v : N \to O$ with $\pi v = \mathrm{id}_N$. Independence of the paths follows by lifting their homotopies in the simply connected plane. The image of this section is open and closed: on a connected evenly covered neighborhood it is exactly one sheet, and all other sheets form its open complement.

Connectedness of O makes it all of O, so $v\pi = \mathrm{id}_O$ as well. Every fiber then has one point, and freeness forces $Q = \{1\}$, contrary to the specified nontriviality.

The primary statement [1, Theorem 1.1, p. 1] lists the five exceptional pairs as the plane with empty fixed set, the plane with one fixed point, the open annulus with empty fixed set, the half-open annulus with empty fixed set, and the punctured projective plane with empty fixed set. For $(N, \{p\})$ and nontrivial Q, the empty-fixed-set pairs do not apply and the remaining plane pair has just been excluded. Thus the stated identity component is contractible, including every nonorientable quotient allowed here. Choose a contraction C_s of that component, with $C_0(b) = b$ and $C_1(b) = c$ constant. In the exact group topology established above,

$$B_{u,t} = C_{1-t}(b_u)\, C_{1-t}(1)^{-1}, \qquad B_{u,0} = 1, \quad B_{u,1} = b_u, \quad B_{u_0,t} = 1, \quad B_{u,t}(p) = p. \tag{IFR.7}$$

The inverse is $C_{1-t}(1)C_{1-t}(b_u)^{-1}$. Both depend continuously on (u, t) in compact-open convergence.

For $Q = \{1\}$, the quotient map π is the open bijection from O to O/Q, with its continuous inverse. We construct the needed B directly in the original disk. Choose a disk coordinate $\iota_0 : \overline{D} \to \overline{\mathbb{D}}$ and put $a = \iota_0(x)$. The maps

$$m_a(z) = \frac{z - a}{1 - \overline{a}z}, \qquad m_a^{-1}(w) = \frac{w + a}{1 + \overline{a}w}, \qquad \iota = m_a \iota_0 \tag{IFR.8}$$

are inverse disk-coordinate adjustments, with $\iota(x) = 0$. For $f_u = \iota h_u \iota^{-1}$ define

$$A_t(f)(z) = \begin{cases} tf(z/t), & 0 < t \le 1,\ |z| \le t, \\ z, & |z| \ge t, \end{cases} \qquad A_0(f) = \mathrm{id}, \qquad A_t(f)^{-1} = A_t(f^{-1}). \tag{IFR.9}$$

The two formulas agree at $|z| = t$ because f fixes the unit circle. They give inverse homeomorphisms fixing that circle and 0. For $t > 0$ the closed-domain pasting in (u, t, z) proves joint continuity for maps and inverses. At $t = 0$ each uniform Euclidean displacement is at most $2t$, independently of f, so their sum is at most $4t$. Uniform continuity of ι^{-1} transfers this estimate to the original metric d. Consequently $\iota^{-1} A_t(f_u)\iota$ is an actual continuous based family from 1 to h_u. Descend it by the explicit maps π and π^{-1} to obtain $B_{u,t}$ satisfying all of (IFR.7). The pointed-plane exception has thus been handled by these original boundary-fixed maps, without contracting the larger ordinary pointed plane group.

The supported correction, its lift, and the original circle. Apply (IFR.3)–(IFR.6) to $M = N$, $K = \{p\}$ and $e_{u,t} = B_{u,t}|_X$. Choose U as in that construction, so $\overline{U}$ is compact. Let $g_{u,t}$ be the resulting family. Its path from the identity fixes p, so its unique pointed lift from C.92 is defined. Put

$$\begin{gathered} G_{u,t}|_O = \lambda(g_{u,t}), \qquad G_{u,t}^{-1}|_O = \lambda(g_{u,t}^{-1}), \\ \pi G_{u,t} = g_{u,t}\pi, \qquad G_{u,t}(x) = x, \qquad G_{u,t}q = qG_{u,t} \quad (q \in Q), \\ G_{u,0} = G_{u_0,t} = 1, \qquad G_{u,t}|_{\pi^{-1}(N \setminus U)} = \mathrm{id}\,. \end{gathered} \tag{IFR.10}$$

The two lift formulas are inverse by uniqueness of the pointed lift of the identity. They are jointly continuous on the interior for maps and inverses by C.92. The same uniqueness retains each original q, rather than allowing an automorphism of the deck group. For a point over $N \setminus U$, its track lifts the constant base track and starts at that point; it is therefore constant. This proves the last equation, and applies to the inverse family.

Here is the quantitative original-circle extension supplied by C.81. Choose a compact $C \subset O$ with $\pi C = \overline{U}$ using finitely many local covering sections. The compact set

$$C' = C \cup \{G_{u,t}(z), G_{u,t}^{-1}(z) : u \in P,\ t \in [0, 1],\ z \in C\} \tag{IFR.11}$$

has one common original source bound by (C.81.8)–(C.81.9). Explicitly, choose compact subsets S, T of the given transitive boundary-fixed centralizer subgroup, with $C \subset S(x)$ and $C' \subset T(x)$.

Set $\mathcal{P} = TS^{-1} \cup \{1\}$. For $z = qc$ choose $a \in S$, $b \in T$ with $c = ax \in C$ and $G_{u,t}^{\pm 1}(c) = bx \in C'$. Then

$$G_{u,t}^{\pm 1}(z) = qbx = ba^{-1}(z), \qquad G_{u,t}^{\pm 1}(z) \in \mathcal{P}(z) \quad (z \in O). \tag{IFR.12}$$

The membership outside $\pi^{-1}\overline{U}$ uses the identity. For all $a \geq 0$ define

$$\omega_{\mathcal{P}}(a) = \max\{d(fz, fw) : f \in \mathcal{P},\ z, w \in \overline{D},\ d(z, w) \leq a\}. \tag{IFR.13}$$

Compactness makes this finite and gives $\omega_{\mathcal{P}}(a) \to 0$ as $a \to 0$. Every $f \in \mathcal{P}$ fixes the original circle. Thus for every ξ on that circle, (IFR.12) gives $d(G_{u,t}^{\pm 1} z, \xi) \leq \omega_{\mathcal{P}}(d(z, \xi))$. Extend both maps by the identity there. This proves their joint circle continuity uniformly in (u, t), and their inverse identities continue to the circle. Joint continuity on the compact disk is exactly continuity of this compact parameter family in both uniform metrics. Therefore every $G_{u,t}$ belongs to the actual $\mathscr{F}$.

Now define H by (IFR.1). Its inverse is the displayed product; both are continuous in the original topology, fix the original circle and x, and commute with every original q. To prove the endpoint assertion with its deck labels, retain the ordinary residual family

$$k_{u,t} = g_{u,t}^{-1} B_{u,t}, \qquad k_{u,t}^{-1} = B_{u,t}^{-1} g_{u,t}, \qquad k_{u,0} = 1, \qquad k_{u,t}|_X = \mathrm{id}\,. \tag{IFR.14}$$

The restriction equation is exactly $g_{u,t}|_X = B_{u,t}|_X$. For each $z \in \pi^{-1}X$, the track $\lambda(k_{u,t})(z)$ lifts the constant base track and starts at z; hence it stays z. Uniqueness of pointed lifts gives the exact group identity $\lambda(g^{-1}b) = \lambda(g)^{-1}\lambda(b)$. Since $\lambda(b_u) = h_u|_O$, its value at $t = 1$ is precisely $H_{u,1}|_O$. This proves $H_{u,1}|_{\pi^{-1}X} = \mathrm{id}$ even when X has several components. Formula (IFR.14) also gives the claimed ordinary isotopy relative to X of the terminal descent. The intermediate family B has only been used on the specified compact embedding trace and in this deck-label argument; its full lift has not been substituted for the actual deformation H.

Exact carriers and both original-metric bounds. Put

$$K_U = \overline{U}, \qquad K_U^- = \bigcup_{u \in P} b_u^{-1}(K_U). \tag{IFR.15}$$

The latter is compact by joint inverse evaluation on $P \times K_U$. The explicit products give $H_{u,t}(z) = h_u(z)$ off $\pi^{-1}K_U^-$ and $H_{u,t}^{-1}(z) = h_u^{-1}(z)$ off $\pi^{-1}K_U$. These are assertions about the actual maps and their input points.

For $y = \pi z$ define the continuous compactly supported functions

$$\begin{gathered} \gamma^{\pm}(y) = \max_{u,t} d_Q(G_{u,t}^{\pm 1} z, z), \qquad \beta^{+}(y) = \max_{u,t} d_Q(G_{u,t}^{-1} h_u z, h_u z), \\ \operatorname{supp} \gamma^{\pm} \subset K_U, \qquad \operatorname{supp} \beta^{+} \subset K_U^-. \end{gathered} \tag{IFR.16}$$

Equivariance proves independence of z; joint continuity and compact maxima prove continuity, using the original topology of d_Q from C.81. The exact support statements follow from (IFR.10) and (IFR.15). Let $a_h^{\pm}(y) = \max_u d_Q(h_u^{\pm 1} z, z)$ be the actual envelopes of CFT, and define for every $a \geq 0$ the common inverse modulus

$$\Omega(a) = \max\{d(h_u^{-1} z, h_u^{-1} w) : u \in P,\ z, w \in \overline{D},\ d(z, w) \leq a\}. \tag{IFR.17}$$

It tends to zero at zero by compactness. Applying this same modulus after every q and using commutation gives $d_Q(h_u^{-1} z, h_u^{-1} w) \leq \Omega(d_Q(z, w))$. The triangle inequality now proves the simultaneous envelope estimates

$$a_H^{+}(y) \leq a_h^{+}(y) + \beta^{+}(y), \qquad a_H^{-}(y) \leq a_h^{-}(y) + \Omega(\gamma^{+}(y)), \tag{IFR.18}$$

where the new envelopes take the maximum over (u, t). Their continuity follows from that of the actual compact family. The displayed bounds and compact supports prove the common C_0 tails. No continuity of the monotone modulus Ω away from zero is needed.

There is also a uniform bound in the exact original sum metric. Put $s_{u,t} = \sup_{z \in \overline{D}} d(G_{u,t}z, z)$. Substitution $z = G_{u,t}w$ and symmetry of d show that the same number is the displacement supremum of $G_{u,t}^{-1}$. Substitution $w = h_u z$ makes the forward change in (IFR.1) exactly $s_{u,t}$; (IFR.17) bounds the inverse change. Hence

$$D_d(H_{u,t}, h_u) \leq s_{u,t} + \Omega(s_{u,t}). \tag{IFR.19}$$

All these quantities use the original closed-disk metric and the actual lifts; no interior conjugacy is used to measure a boundary tail.

Compatible finite nested-core stages and the remaining estimate. Choose a finite nested chain $X_1 \subset \cdots \subset X_r$ of compact subpolyhedra, with X_1 a closed disk neighborhood of p. The first stage above produces an actual endpoint fixing $\pi^{-1}X_1$, and (IFR.14) places its descents in $\mathrm{Homeo}_{X_1}(N)_0$. For each later stage this component is contractible by the same Theorem 1.1: its nonempty fixed set contains a disk, so none of the five listed exceptional pairs applies, for any orientation of N. Normalize that contraction by (IFR.7), now inside the group fixing X_i, and use (IFR.3)–(IFR.6) with $K = X_i$ and $X = X_{i+1}$.

Each correction fixes X_i throughout its path from the identity. Its actual lift therefore fixes every point of $\pi^{-1}X_i$ throughout, by the same constant-track argument. All earlier fixed points stay fixed during the next actual deformation. Equation (IFR.14) supplies the relative component witness for the next endpoint. Induction gives all stages and their explicitly inverse maps; finite concatenation is continuous in the original topology at the matching endpoints. This construction includes compact polyhedra with edges, cycles and complementary components through their actual embeddings.

For each finite stage, (IFR.15)–(IFR.19) prove its carriers and its original map-and-inverse control. They do not bound $s_{u,t}$ by the initial displacement envelope outside the already fixed core. In particular no estimate forcing these correction sizes to tend to zero along an exhaustion has been proved here. The remaining global task is that quantitative estimate for a compatible infinite sequence in the original topology. The present result is the constructed common fixed region and the compatible finite stages just proved.

References

[1] Tatsuhiko Yagasaki. Homotopy types of homeomorphism groups of noncompact 2-manifolds. arXiv:math/0010223, 2000. Verified: supplied primary TeX source opened. Theorem 1.1, the definitions of compact-open homeomorphism groups and fixed compact subpolyhedra, Theorem 2.1 on local extension, and Lemma 2.1 on compact surfaces read. Lemma 2.1 supplies the needed compact marked-surface homotopy triviality; its cited Hamstrom and Scott originals are not independently claimed to have been opened.